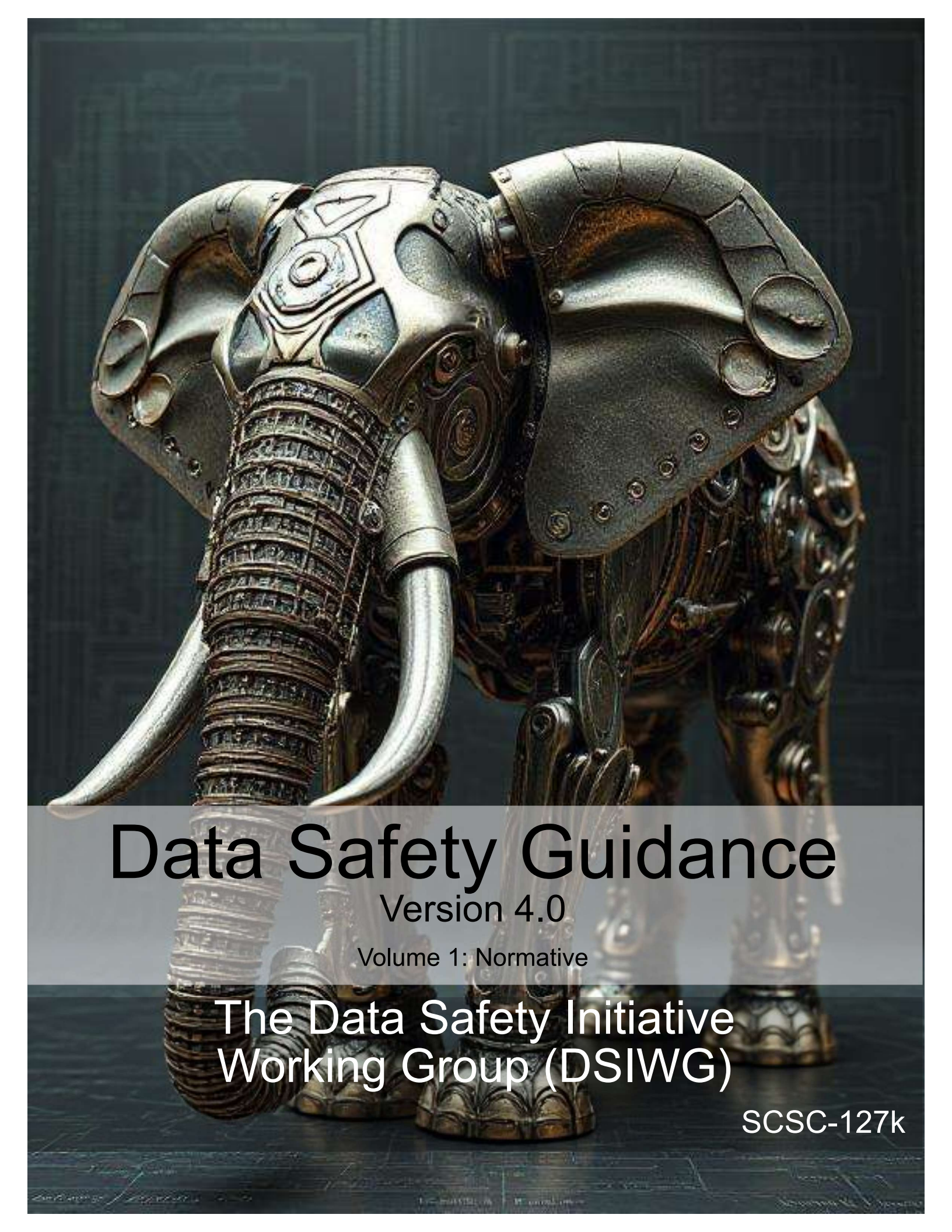




Data Safety Guidance

Version 4.0

Volume 1: Normative

The Data Safety Initiative
Working Group (DSIWG)

SCSC-127k

SCSC Publication Number: SCSC-127J

This work is licensed under the Creative Commons Attribution 4.0 International License. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/> or send a letter to Creative Commons, PO Box 1866, Mountain View, CA 94042, USA. You are free to share the material in any form and adapt the material for any purpose providing you attribute the material to the [Safety-Critical Systems Club \(SCSC\) Data Safety Initiative Working Group](#), reference the source material, include the licence details above, and indicate if any changes were made. See the license for full details.

This document was prepared using the $\text{\LaTeX}2_{\epsilon}$ typesetting system.

Editing and typesetting by Mark Templeton and Tim Rowe.

Cover design by DeepAI.org and Tim Rowe, based on an original by Paul Hampton.

The [Safety-Critical Systems Club \(SCSC\)](#) is the professional network for sharing knowledge regarding safety-critical systems. It brings together:

- engineers and specialists from a range of disciplines working on safety-critical systems in a wide variety of industries;
- academics researching the arena of safety-critical systems;
- providers of the tools and services that are needed to develop the systems; and
- the regulators who oversee safety.

Through publications, seminars, workshops, tutorials, a web site and, most importantly, at the annual [Safety-critical Systems Symposium \(SSS\)](#), it provides opportunities for these people to network and benefit from each other's experience in working hard at the accidents that don't happen. It focuses on current and emerging practices in safety engineering, software engineering, and product and process safety standards.

This document was written by the [Data Safety Initiative Working Group \(DSIWG\)](#), which is convened under the auspices of the [SCSC](#). The document supports the [DSIWG's](#) vision, which is to have clear guidance that reflects emerging best practice on how data (as distinct from software and hardware) should be managed in a safety-related context. This update takes account of the consensus that a process-based guidance document will complement existing safety management processes, making it more usable. It was formally released at [SSS'26](#), 10–12 February 2026, details of which may be found at <https://scsc.uk/events-diary/safety-critical-systems-club-symposia/safety-critical-systems-symposium-2026>

Comments on this document are actively encouraged. These can be emailed to:

comments@data-safety.scsc.uk.

While the authors and the publishers have used reasonable endeavours to ensure that the information and guidance given in this work is correct, all parties must rely on their own skill and judgement when making use of this work and obtain professional or specialist advice before taking, or refraining from, any action on the basis of the content of this work. Neither the authors nor the publishers make any representations or warranties of any kind, express or implied, about the completeness, accuracy, reliability, suitability or availability with respect to such information and guidance for any purpose, and they will not be liable for any loss or damage including without limitation, indirect or consequential loss or damage, or any loss or damage whatsoever (including as a result of negligence) arising out of, or in connection with, the use of this work. The views and opinions expressed in this publication are those of the authors and do not necessarily reflect those of their employers, the [SCSC](#) or other organizations.

Data Safety Guidance

The Data Safety Initiative Working Group [DSIWG]

February 2026

Change History

Version	By	Status	Date
1.0	The DSIWG Team	First draft for external review	31-JAN-2014
1.1	The DSIWG Team	(Internal edition for DSIWG use only)	09-DEC-2014
1.2	The DSIWG Team	For publication at SSS'15	23-JAN-2015
1.3	The DSIWG Team	For publication at SSS'16	29-JAN-2016
2.0	The DSIWG Team	For publication at SSS'17	30-JAN-2017
3.0	The DSIWG Team	For publication at SSS'18	26-JAN-2018
3.1	The DSIWG Team	For publication at SSS'19	01-FEB-2019
3.2	The DSIWG Team	For publication at SSS'20	11-FEB-2020
3.3	The DSIWG Team	For publication at SSS'21	09-FEB-2021
3.4	The DSIWG Team	For publication at SSS'22	08-FEB-2022
3.5	The DSIWG Team	For publication at SSS'23	07-FEB-2023
3.6	The DSIWG Team	For publication at SSS'24	13-FEB-2024
3.7	The DSIWG Team	For publication at SSS'25	04-FEB-2025
4.0	The DSIWG Team	For publication at SSS'26	10-FEB-2026

Changes Since the Last Edition

This is a major rewrite of the guidance. The document has been split into three volumes , each giving progressively more detail.

The process remains substantially the same. The intention is that compliance with this version does imply compliance with earlier versions [???Need to check this is still true](#), but some material that was informative in earlier versions has been made normative, so compliance with earlier versions of the guidance does not guarantee compliance with this version.

Future work

[???Update future work](#) [???Consider whether future work needs to be included in all parts](#) MCA Ltd has continued to work with the [DSIWG](#) to develop a prototype software tool to assist in the automation of the processes described in this guidance document. A working version of the tool has been developed and organizations that could benefit from the use and further development of the tool are urged to contact MCA at [Mission Critical Applications Limited \(radish@mca-ltd.com\)](mailto:radish@mca-ltd.com).

A number of improvements to the guidance are currently planned. These improvements are intended to clarify the application of the data safety process [???Review future work](#) and include:

- further detail on the assurance of communications and data flows,
- data safety considerations associated with distributed [datasets](#) and Blockchain,
- addition of new [treatments](#) to the tables in [??](#),
- review of the tables of [treatments](#), with the aim of making them easier to use,

- further explanation of some [treatments](#), where their use or benefit is not immediately apparent,
- reordering of parts of the document to improve readability, especially as regards likelihood,
- further detail on tool assurance,
- harmonisation of language and guidance on how organizations may expand the tables to incorporate their own internal processes.
- guidance on the application of the data safety culture questionnaire,

Several of these changes are likely to cause parts of the document to be re-ordered – they have therefore been deferred to the next major update, in version 4.0 of the guidance.

If you or your organization are interested in learning more about the work of the [DSIWG](#) or joining any of the sub-groups, please visit the [SCSC](#) website, where more information including contact details may be found on the “Working groups” section of the site.

Related working groups

The [SCSC](#) sponsors initiatives to develop methods and techniques through a number of working groups. These groups each address safety aspects peculiar to their domain, including data aspects when appropriate. The current list of working groups includes:

- Assurance Cases,
- Security Informed Safety,
- Safe AI Working Group,
- Safer Complex Systems.

This page is intentionally blank

Foreword

Data is here. Data is growing. Data is causing harm.

Data is here: Data is becoming ever more important in our lives: influencing, managing and even controlling many critical aspects. The use of [artificial intelligence \(AI\)](#) systems is a new, exciting, but potentially hazardous use of data. [Large language model \(LLM\)](#) based systems are trained on vast amounts of data, and it is this data which enables them to be useful.

Some of this data is related to our personal safety and well-being. Consider, for example, the importance of data defining the layout of railway signals, data that indicates the position of underwater obstructions in nautical channels or data that is used to train a vision recognition system to detect tumours in medical images. Organizations now make significant decisions (including safety-related decisions) based solely on data held in systems. Hence, organizations need to safely manage, control and process their data. In particular, they must actively manage key [data properties](#) that preserve safety.

Data is growing: There are at least two reasons why the use of data has grown and, equally important, why it is expected to continue to grow. The first relates to the rapid expansion of the area loosely termed “Big Data”, including the use of large [datasets](#) to support machine learning and [AI](#) applications. The second is the growing use of systems of systems, where data is the lifeblood that connects together disparate elements and allows a cohesive capability to be built. Put simply, the need to address data-related issues is a pressing problem and will continue to be so.

Data is causing harm: Strictly speaking, data can neither cause nor prevent harm. However, mistakes in data, or the inappropriate use of data, within safety-related systems have been factors in a number of documented accidents and incidents. Examples include aircraft attempting to take off from the wrong runway (and consequently crashing), ships running aground, and patients being exposed to higher than planned doses of radiation.

Against this background, the [DSIWG](#) was established under the auspices of the [SCSC](#). The [DSIWG](#)’s aim is to develop clear, cross-sector guidance that reflects emerging best practice on how data (as opposed to software or hardware) should be managed in a safety-related context. For the most part, this guidance is based on well-established techniques, and it has been designed to be compatible with current safety standards and to integrate with existing safety management systems. What is new, however, is the explicit and relentless focus on data, making it a “first-class citizen” within system safety analyses. Because of this focus, this guidance should help organizations identify, analyse, evaluate and treat data-related risks, thus reducing the likelihood of data-related issues causing harm in the future.

Quick Start Guide

???Revise quick start guide for new structure

Data really powers everything that we do.
Jeff Weiner

This section provides a single-page introduction to [data safety guidance \(DSG\)](#). For first-time readers this should help place individual sections within an appropriate context. It should also help returning readers quickly navigate the document's contents.

Systems are changing. The role of data is becoming more prominent. Hence, data needs to be considered as a “first-class citizen” in system safety analyses. This will help mitigate organizational and system-level risks associated with the use of data.

The guidance is in three volumes, published as separate documents.

Volume 1 (this volume) contains the normative material. It describes the “what” of the guidance and states what is needed to ensure conformance with this guidance, if required. It should be of use to those establishing processes and procedures for data safety assurance and for those doing the actual assurance;

Volume 2 contains the informative material. It describes the “how” of the guidance and should be of particular value to data safety practitioners;

Volume 3 contains the discursive material. It describes the “why” of the guidance and gives background and supporting material intended to improve practitioner's understanding of the data safety management processes and of issues relating to data safety management.

A data safety management process has been developed. This is based on four phases:

- establish context;
- identify risks;
- analyse risks; and
- evaluate and [treat](#) risks.

???This part needs rewriting when structure is stable

- The underlying principles of the guidance are described in [Chapter 2](#).
- The objectives and outputs to satisfy the principles are described in [Chapter 3](#).
- Normative definitions and abbreviations are described in [Chapter 4](#).
- The activities of each phase (and associated [tailoring information](#)) are described in ??.
- Additional guidance [information](#) for each phase is described in ??.
- A worked example is provided in ??.

- A collection of appendices provide more detail, including:
 - A discussion illustrating how the underlying principles link to the objectives (Appendix ??);
 - An [organizational data risk \(ODR\)](#) assessment questionnaire (Appendix ??);
 - A data safety culture questionnaire (Appendix ??);
 - A questionnaire to help assess the data maturity of a supplier (Appendix ??);
 - A list of data categories (Appendix ??);
 - A collection of [hazard and operability study \(HAZOP\)](#) guidewords (Appendix ??);
 - The suggested contents of a [data safety management plan \(DSMP\)](#) (Appendix ??);
 - A summary of accidents and incidents in which data was potentially a causal factor (Appendix ??);
 - A discussion of topics loosely related to system lifecycles (Appendix ??);
 - Considerations regarding [machine learning \(ML\)](#) (Appendix ??);
 - A discussion of the risks of AI and autonomy (Appendix ??);
 - An introduction to the concepts of both dark and dazzle data (Appendix ??);
 - The concepts of black swan, dragon king, perfect storm and Pudding Lane data (Appendix ??);
 - Considerations for the assurance and qualification of data-handling tools (Appendix ??);
 - An introduction to the RADISH tool, that has been developed to assist in the application of the guidance within this document (Appendix ??);
 - Issues that may arise when migrating, porting, importing or exporting data (Appendix ??);
 - Some of the data issues that made management of the Covid-19 virus difficult (Appendix ??);
 - Examples of ways that [data safety assurance levels \(DSALs\)](#) may be customised, with particular focus on likelihood (Appendix ??);
 - Lists of acronyms, definitions and glossary entries ([Appendix 5](#)); and
 - A collection of references ([Appendix 6](#)).

When reference is made to the content of other volumes, this is indicated by a right-arrow and the volume number shown in bold text. So “Chapter → **3:6**” refers to chapter 6 in volume 3.??? I didn't like the “Volume **V:x**” format. This is the format used by The LaTeX companion. What do people think?

This page is intentionally blank



Contents

1	Introduction (Informative)	1
1.1	Aim and Scope	1
1.2	Structure of the Guidance Documents	1
1.3	Intended Relationship to Other Documents	2
1.4	Normative, Informative and Discursive Text	3
1.5	Conformance	3
2	Principles (Normative)	5
2.1	Principle 1: data safety requirements shall be defined to address the data contribution to system hazards	5
2.2	Principle 2: the intent of the data safety requirements shall be maintained throughout requirements decomposition	5
2.3	Principle 3: data safety requirements shall be satisfied	5
2.4	Principle 4: Hazardous system behaviour arising from the system's use of data shall be identified and mitigated	6
2.5	Principle 5: The confidence established in addressing the data safety assurance principles shall be commensurate to the contribution of the data to system risk	6
3	Objectives (Informative)	7
3.1	General	7
3.2	Principle 1	7
3.3	Principle 2	8
3.4	Principle 3	9
3.5	Principle 4	10
3.6	Principle 5	10
3.7	Principle 6	11
4	Definitions (Normative)	13
5	Acronyms, Definitions and Glossary (Discursive)	15
5.1	Acronyms	15
5.2	Definitions and Glossary	15

6	References (Discursive)	19
	Index	21

List of Tables

3.1	P1 Objectives and Outputs	7
3.1	P1 Objectives and Outputs (continued)	8
3.2	P2 Objectives and Outputs	8
3.2	P2 Objectives and Outputs (continued)	9
3.3	P3 Objectives and Outputs	9
3.4	P4 Objectives and Outputs	10
3.5	P5 Objectives and Outputs	10
3.5	P5 Objectives and Outputs (continued)	11
3.6	P6 Objectives and Outputs	11

This page is intentionally blank

List of Figures

This page is intentionally blank

1 Introduction (Informative)

???consider whether the introduction needs to be included in all parts (normative, informative and discursive, perhaps and shall, should, may, but they should be in definitions anyway)

We're entering a new world in which data may be more important than software.

Tim O'Reilly

1.1 Aim and Scope

This guidance document aims to:

- describe the data safety problem;
- provide methods for identifying and analysing levels of risk; and
- recommend methods and approaches for evaluating and treating those risks.

It has been written for a wide readership. Its target audience is all those who have an interest in or a responsibility for safety-related data within systems, including managers, developers, safety engineers, assurers (including independent safety auditors), regulators, and operators.

The document is also intended to cover a number of different sectors. It identifies a wide spectrum of safety-related data that exists in many forms within systems, from specification and requirements data to maintenance and disposal data, and everything in between. In particular, this document is not just concerned with numerical or well-structured data used during system operation.

The contents of the document are considered mature enough to be useful, but represent current thoughts on what is a complex and evolving area. Furthermore, to allow it to be produced within a reasonable timescale, this edition focuses on key items. It is not intended to be exhaustive. For example, this guidance document does not consider issues relating to staff competence or organizational structure.

There are various understandings of the word “[safety](#)”, from those limited to physical harm to the person: “The state of being protected from or guarded against hurt or injury; freedom from danger” ([Oxford English Dictionary \(OED\)](#)) to those that consider more general harm: “Freedom from conditions that can cause death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment.” ([1]). This guidance concentrates on the former but is applicable more widely, and from time to time other types of harm will be considered, particularly but not exclusively in the examples of incidents and accidents (Chapter **3:6**).

1.2 Structure of the Guidance Documents

The guidance is in three parts, published as separate documents.

Part 1 contains the normative material. It describes the “what” of the guidance and states what is needed to ensure conformance with this guidance, if required. It should be of use to those establishing processes and procedures for data safety assurance and for those doing the actual assurance work;

Part 2 contains the informative material. It describes the “how” of the guidance and should be of particular value to data safety practitioners;

Part 3 contains the discursive material. It describes the “why” of the guidance and gives background and supporting material intended to improve practitioner’s understanding of the data safety management processes and of issues relating to data safety management.

1.3 Intended Relationship to Other Documents

This document is intended to be used as a supplement to existing standards and norms that are relevant to the scope of the work being undertaken. It may be used to provide a deeper insight into the risks that data poses to the project team’s outputs, allowing them to produce credible improvements to the safety argument. Where a standard or norm sets out specific data-related objectives then, unless agreed otherwise with the regulator or safety duty holder, they shall take precedence over the guidance provided herein.

In the longer term, the hope is that future standards and norms will take up relevant concepts, approaches and methods from those in this document. The [DSIWG](#) also hopes that organizations will include the concepts, approaches and methods in their own safety management processes.

1.4 Normative, Informative and Discursive Text

Consider whether this is relevant to Part 2 Three types of text are used within this guidance document:

Normative text, which is prescriptive. Typically, this text is restricted to describing objectives and outputs.

Informative text, which is descriptive text that is closely linked to the normative text. Typically, this text provides a suggested way by which conformance with the normative text may be achieved, but alternative means of conformance are possible.

Discursive text, which contains discussions that are relevant to the general topic of data safety, but which are not closely linked to the normative text. A discussion on the relationship between data and software is an example of such text. Descriptions of historical incidents and accidents are another.

Each section and appendix of this guidance document contains a single text type. The relevant type is indicated in the section or appendix title.

1.5 Conformance

There may be occasions when it is desirable or necessary to make a claim of conformance to the objectives listed in this document. Such a claim may be required, for example, if this document is explicitly included as a normative reference from a formal standard. Alternatively, it may be required as part of an organization's internal processes.

To facilitate conformance claims, the following terminology is used within the normative parts of this guidance document:

SHALL denotes items where evidence of conformance must be provided in order to claim conformance with this guidance document.

SHOULD denotes items where, in some circumstances, there may be valid reasons for not conforming to a particular item. The full implications of non-conformance must be understood, documented and approved in order to claim conformance with this guidance document.

MAY denotes items that are optional. These may be advantageous in some circumstances but not in others. Organizations are free to adopt any approach to these items without the need for further justification.

The terms have their normal English meanings in discursive and descriptive sections.

??extract principles

2 Principles (Normative)

Errors using inadequate data are much less than those using no data at all.
Charles Babbage

Hawkins *et al.* established some generic software safety assurance principles, which are commonly referred to as “4 + 1” [2]. Given the close links between software and data it is helpful to consider these principles from a data safety assurance perspective. The results are detailed below, with each principle being considered in turn.

2.1 Principle 1: data safety requirements shall be defined to address the data contribution to system hazards

Data pervades active system operation, as well as the system’s specification, realisation, [verification](#), [validation](#), certification, training, maintenance, and retirement. Moreover, data may be passed from one system to another, sometimes over a significant period of time. Data may be assimilated and converted from prior uses into new uses, or simply used as-is, by many systems. It is stored in media whose storage [integrity](#) decays. The system context for [data safety requirements](#) may be specific to a particular system’s or process’s use of the data, or it may be generalised to a class of related systems. Hence [data safety requirements](#) are needed for any safety-related system that interacts with data.

2.2 Principle 2: the intent of the data safety requirements shall be maintained throughout requirements decomposition

[Data safety requirements](#) establish the system’s safety properties for data, for the system’s use of data, for the management of data, and for the engineering lifecycle of both the system and its associated data. The system’s requirements hierarchy must preserve the intent of the [data safety requirements](#) (and hence the system’s safety-related [data properties](#)). Moreover, the applied engineering process for both the system’s realisation and subsequent lifecycle stages shall demonstrate that the data safety properties are preserved.

2.3 Principle 3: data safety requirements shall be satisfied

Evidence is required that the system satisfies all of the [data safety requirements](#) imposed on it for all anticipated operating conditions. Moreover, the [data safety requirements](#) that pertain to the data’s lifecycle outside of the system shall be evidentially demonstrated prior to the system acting on such data, or the system shall be able to adequately defend against unsatisfied [data safety requirements](#). In other words, either the data can be shown to demonstrate the required [data properties](#) prior to being used or the system can implement adequate defences and [mitigations](#) against data that does not conform to the required safety properties.

2.4 Principle 4: Hazardous system behaviour arising from the system's use of data shall be identified and mitigated

This is an intentionally broad statement because data is conceptual and not physical; it is the contextualised use of data that could result in a system hazard. Data safety assurance principle 1 deals with system-level hazards arising from data, whereas Data safety assurance principle 4 is concerned with hazards that arise from the way the system uses its data, that is, whether the system's design and implementation introduce further hazards. An example is a ship navigation system's display of hydrographic chart data, where a wide field display results in small features disappearing (due to image scale) when it is critical that situational awareness of such hazards is maintained. The associated objectives and outputs are shown in Part 2 Chapter ??.

2.5 Principle 5: The confidence established in addressing the data safety assurance principles shall be commensurate to the contribution of the data to system risk

(Called Principle 4+1 by Hawkins.) The confidence in the evidence that demonstrates establishment of the first four Data Safety Assurance Principles shall be proportionate to the contribution data (or a particular data artefact) makes to the system hazards. ???expand dark data, make into objective ???Add definitions for cygnology types ???Identify process for dark data

3 Objectives (Informative)

I'm a bit of a freak for evidence-based analysis. I strongly believe in data.
Gus O'Donnell

3.1 General

The data safety assurance principles (Chapter 2) provide the underpinning philosophy for this guidance. An implementation of the guidance would be based around these objectives. This section identifies the objectives to satisfy the principles.

3.2 Principle 1

Data safety requirements shall be defined to address the data contribution to system hazards.

This guidance uses a metric called [data safety assurance level \(DSAL\)](#) for this purpose., described in more detail in Part 2:6 and Part 2:8. The objectives and outputs supporting this principle are shown in Table 3.1.

Table 3.1: P1 Objectives and Outputs

Id	Objectives	Outputs	Reference
1.a	System context and intended use SHALL be established.	A description of the system and its intended use e.g. in a Concept of Operation (CONOPS) . This SHOULD include an estimate of the level of data risks.	
1.b	Legal and regulatory obligations and requirements SHALL be incorporated.	A record of all applicable laws and regulations.	
1.c	Applicable standards and relevant guidance SHALL be incorporated	A record of all applicable standards and relevant guidance.	
1.d	Existing data quality and data assurance requirements and constraints SHALL be identified	A record of applicable existing quality and data assurance requirements and constraints.	
1.e	Data safety artefacts SHALL be identified	A collection of data artefacts, described at an appropriate level of detail.	
1.f	Risks SHALL be identified and linked to data safety artefacts	A list of risks, linked to data artefacts and data properties.	
		A description of the process used for risk identification	
1.g	Data safety activities SHALL be defined and documented	A plan for covering all data safety relevant aspects needs to be created.	
1.h	Key data stakeholders SHALL be identified	A list of key stakeholders for data safety activities, including those affected by the data.	

Continued on next page

Table 3.1: P1 Objectives and Outputs (continued)

Id	Objectives	Outputs	Reference
1.i	Missing, hidden, and obscured data SHALL be considered for safety impact	A plan to address potential relevant missing (Dark Data) and obscured data (Dazzle Data).	
1.j	External interfaces SHALL be defined. This MAY include a list of data owners, linked to data artefacts.	An interface control plan.	
1.k	A complete list of data assurance requirements SHALL be compiled, based on all the legal and regulatory needs, external inputs, known risks, etc.	A complete list of data assurance requirements.	
1.l	Relevant historical data-related accidents and incidents SHOULD be reviewed for additional requirements	A documented review of relevant historical data-related accidents and incidents.	
1.m	DSALs SHALL be established and justified.	A list of DSALs and their justifications.	

3.3 Principle 2

The intent of data safety requirements shall be maintained throughout requirements decomposition.

This principle is based on standard systems engineering practices whereby a system is gradually developed at increasing levels of design detail. In order to cater for a wide range of systems, across a wide range of economic sectors, DSG does not specifically require an explicit hierarchical decomposition of requirements. However, it does note that data artefacts MAY be defined at a number of levels of increasing detail. The objectives and outputs supporting this principle are shown in Table 3.2.

Table 3.2: P2 Objectives and Outputs

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
2.a	Levels of abstraction of the data SHALL be defined.	Levels of Abstraction report. This MAY be in diagrammatic form. There MAY be several levels of data abstraction, presenting different relevant perspectives.	
2.b	DSALs SHALL be flowed through levels of the design	Traceability records showing the flow of DSALs through the design.	
2.c	Data assurance requirements SHALL be established for each level of system design.	Requirements documents.	
2.d	Data assurance requirements SHALL be traced between each level of system design	Traceability report	
2.e	Data-related interactions between components SHALL be considered.	Interactions report.	
2.f	Data-related interactions with other systems SHALL be considered.	Interactions report.	

Continued on next page

Table 3.2: P2 Objectives and Outputs (continued)

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
2.g	Techniques and approaches for mitigating data risks SHALL be identified	Mitigation report.	
2.h	Unintended interactions affecting data SHALL be identified.	Interactions report.	
2.i	Techniques and tools used for data analysis SHALL be justified.	Technique and tool justification.	

3.4 Principle 3

Data safety requirements shall be satisfied.

The objectives and outputs supporting this principle are shown in Table 3.3.

Table 3.3: P3 Objectives and Outputs

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
3.a	Methods and techniques used to verify data safety requirements SHALL be documented and justified	A record of the treatment adopted for each of the identified risks	
3.b	Data safety requirements SHALL be verified.	Evidence that the treatment has been successfully implemented E.g. through test, analysis, review or other methods	
		Verification reports	
3.c	Methods and tools used to provide supplemental and supporting assurance SHALL be documented.		
3.d	DSALs SHALL be satisfied.		
3.e	Trusted tools SHALL be used to produce verification evidence		
3.f	Verification evidence SHOULD include evidence acquired from field or historical usage where available		
3.g	Conformance with data safety requirements SHALL be documented.	Conformance matrix	
		An assessment as to whether the risk has been suitably mitigated (and, if not, plans for further mitigation activities).	
3.h	Data safety requirements which are partially or not met SHALL identified and justified		
3.i	Data which is missing or hidden SHALL be assessed for impact.	A missing and hidden data resilience report.	

3.5 Principle 4

Hazardous system behaviour arising from the system's use of data shall be identified and mitigated.

The objectives and outputs supporting this principle are shown in Table 3.4. **consider whether it's appropriate to refer to Black swan data, etc, which are not normatively defined.**

Table 3.4: P4 Objectives and Outputs

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
4.a	Key properties of the data SHALL be analysed for failures which lead to safety risks.	Data-focussed Functional Failure Analysis.	
4.b	Unintended behaviour resulting from data SHALL be identified and analysed. The analysis SHOULD include black swan data , dragon king data , perfect storm data , and Pudding Lane data .	Unintended Behaviours Report.	
4.c	reduced or absent data SHALL be identified and analysed.	Unintended Behaviours Report.	
4.d	Emergent and unanticipated effects due to data SHALL be investigated.		
4.e	Resilience measures to deal with unknowns of the data SHALL be considered.		
4.f	Unintended behaviour resulting from malicious data SHALL be identified and analysed.		

3.6 Principle 5

The confidence established in addressing the data safety assurance principles shall be commensurate to the contribution of data to system risk.

The objectives and outputs supporting this principle are shown in Table 3.5.

Table 3.5: P5 Objectives and Outputs

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
5.a	DSALs SHALL be demonstrably applied.		
5.b	Assurance artefacts SHALL be produced with a level of rigour commensurate with the DSAL and data safety requirements.		
5.c	Activities, methods, analyses, and tools used to provide assurance SHALL be appropriate for the DSAL and stakeholder group.		

Continued on next page

Table 3.5: P5 Objectives and Outputs (continued)

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
5.d	Specific confidence demands from stakeholder groups SHALL be addressed.		
5.e	Residual risks SHALL be documented.		

3.7 Principle 6

We don't have a Principal 6! The objectives and outputs supporting this principle are shown in Table 3.6.

Table 3.6: P6 Objectives and Outputs

Id	Objectives	Brief Explanation and Deliverables	Reference in the Guidance
6.a	All changes that impact these principles and objectives SHALL be assessed and managed.		
6.b	Assurance artefacts SHALL be maintained.		
6.c	Measures SHALL be in place to monitor the behaviour of the SCS at appropriate intervals throughout its life.		
6.d	Use SHALL be monitored for change and an assurance impact analysis undertaken where necessary.		
6.e	Evolution and incremental change SHALL be considered.		
6.f	Longevity and obsolescence of data SCS SHALL be considered.		
6.g	Lessons learned from successful operation, failures and incidents SHOULD be reviewed		

This page is intentionally blank

4 Definitions (Normative)

I love data. I think it's very important to get it right, and I think it's good to question it.

Mary Meeker

artefact, data

An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

property, data

A characteristic that can be exhibited by a [data artefact](#).

safety assurance level, data

An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

safety requirement, data

A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

treatment

An action taken to reduce or control risk. This might be [mitigation](#) of the risk or elimination of the hazard.

This page is intentionally blank

5 Acronyms, Definitions and Glossary (Discursive)

The plural of anecdote is not data.
Mark Berkoff

5.1 Acronyms

AI	artificial intelligence
CONOPS	Concept of Operation
DSAL	data safety assurance level
DSG	data safety guidance
DSIWG	Data Safety Initiative Working Group
DSMP	data safety management plan
HAZOP	hazard and operability study
LLM	large language model
ML	machine learning
ODR	organizational data risk
OED	Oxford English Dictionary
SCSC	Safety-Critical Systems Club
SSS	Safety-critical Systems Symposium

5.2 Definitions and Glossary

accuracy

Closeness of agreement between a measured quantity value and a true quantity value of a measurand. [3]

artefact, data

(Normative) An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

black swan data

Data that is totally unexpected by those receiving it.

completeness

- Property of having every necessary part or element.
- Completeness of the data provided. RTCA/DO-200A

configuration data

- Data that configures a generic software system to a particular instance of its use. (EC) No 482/2008 [4]
- Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [5]
- Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [5]

consistency, data

The property that the data adheres to a common world view (e.g., units).

correctness, data

self-[consistency](#), protection against alteration or corruption and [consistency](#) with the functional requirements of the [data-driven system](#). IEC 61508 Part 3 [6]

data

- A thing given or granted; something known or assumed as fact, and made the basis of reasoning or calculation; an assumption or premiss from which inferences are drawn. [OED](#)
- A reinterpretable representation of [information](#) in a formalized manner suitable for communication, interpretation or processing. ISO/IEC 2382 [7]

data-driven system

System which relies upon [configuration data](#) or lookup tables to define the functionality of the system. IEC 61508 Part 4 [8]

dataset

Identifiable collection of data. Note that a dataset may be a smaller grouping of data which, though limited by some constraint such as spatial extent or feature type, is located physically within a larger dataset. Theoretically, a dataset may be as small as a single feature or feature attribute contained within a larger dataset. A hardcopy map or chart may be considered a dataset. BS EN ISO 19131:2008 [9]

dragon king data

Data whose effect might have been foreseen, but that leads to an unexpected escalation with major impact

information

- Knowledge communicated concerning some particular fact, subject, or event; that of which one is apprised or told - intelligence, news - as contrasted with data. [OED](#)
- Knowledge that has a contextual meaning. ISO/IEC 2382 [7]

integrity, data

- The assurance that a data element retrieved from a storage system has not been corrupted or altered in any ways since the original data entry or latest authorised amendment. RTCA/DO-200A [10]
- The degree of assurance that a [data item](#) and its value have not been lost or altered since the data origination or authorised amendment. (EU) No 73/2010 [11]
- The degree of undetected (at system level) non-conformity of the input value of the [data item](#) with its output value. (EU) No 1207/2011 [12]
- The property of protecting the [accuracy](#) and [completeness](#) of assets, i.e., that which has value to the organization. ISO 27001:2013 [13]

item, data

Single attribute of a complete [dataset](#), which is allocated a value that defines its current status. (EU) No 73/2010 [11]

mitigation

Steps taken to control or prevent a hazard from causing harm and reduce risk to a tolerable or acceptable level. [14]

owner, data

(Normative) The individual or organization responsible for a particular [data artefact](#) or collection of [data artefacts](#).

perfect storm data

Combinations, sets or occurrences of data (or absences of data) that would never have been thought possible to occur together, and when they do have a large and undesirable impact.

property, data

(Normative) A characteristic that can be exhibited by a [data artefact](#).

Pudding Lane data

Data that is unknowingly and unexpectedly critical to the whole operation and if missing or incorrect in some way, has a dramatic and negative impact. After the event it may be obvious that it was critical.

response

(Normative) The way in which an identified risk is addressed; possible responses include avoid / eliminate, treat, or accept as sufficiently low.

safety

- The state of being protected from or guarded against hurt or injury; freedom from danger. ??
- Freedom from conditions that can cause death, injury, occupational illness, damage to or loss of equipment or property, or damage to the environment. [1]

safety assessment, data

(Normative) The process of explicitly considering data as part of a system safety assessment, via the means of [data artefacts](#), Data Properties and [DSALs](#).

safety assurance level, data

(Normative) An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

safety requirement, data

(Normative) A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

stakeholder

(Normative) An individual or organization that has some relationship to the system, possibly including a power of veto.

tailoring

Adaptation of processes etc. to be appropriate for a specific system and context

treat

To apply a [treatment](#)

treatment

(Normative) An action taken to reduce or control risk.

validation, data

- The activity whereby a data element is checked as having a value that is fully applicable to the identity given to the data element, or a set of data elements that is checked as being acceptable for their purpose. RTCA/DO-200A [10]
- Process of ensuring that data meets the requirements for the specified application or intended use. (EU) No 73/2010 [11]

validity, period of

Period between the date and time on which information becomes available and the date and time on which the [information](#) ceases to be effective. Based on (EU) No 73/2010 [11]

verification, data

Evaluation of the output of a process to ensure [correctness](#) and [consistency](#) with respect to the inputs and applicable data standards, rules and conventions used in that process. Based on (EU) No 73/2010 [11]

The normative list of definitions is at [section 4](#). Normative definitions have been repeated here for convenience.

6 References (Discursive)

Data opportunities multiply as the data is transformed.
Sun Tzu misquoted

Bibliography

- [1] System safety. standard Mil-Std-882E, Department of Defense, 2012.
- [2] R Hawkins, I Habli, and T Kelly. *The Principles of Software Safety Assurance*. International System Safety Society, Boston, Massachusetts, USA, 2013.
- [3] ISO/IEC. Information technology — data governance — data collaboration framework. <https://www.iso.org/obp/ui#iso:std:iso-iec:25642:ed-1:v1:en:term:3.9>. Accessed: 2 October 2025.
- [4] Commission Regulation (EC) No 482/2008 of 30 May 2008 establishing a software safety assurance system, as amended. URL <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32008R0482&qid=1611781633022>, May 2008. Accessed: 27 January 2021, but no longer in force.
- [5] Guidelines for ANS software safety assurance. Standard EUROCAE/ED-153, European Organisation for Civil Aviation Equipment, August 2009. Used for definitions only.
- [6] Functional safety of electrical/electronic/ programmable electronic safety-related systems: Software requirements. Standard BS EN 61508-3:2010, International Electrotechnical Commission, June 2010.
- [7] Information Technology. Vocabulary. Part 1: Fundamental terms. Standard ISO/IEC 2382-1:1993, International Standards Organisation, 1993.
- [8] Functional safety of electrical/electronic/ programmable electronic safety related systems: Definitions and abbreviations. Standard BS EN 61508-4:2010, International Electrotechnical Commission, June 2010.
- [9] Geographic Information. Data Product Specifications. Standard BS EN ISO 19131:2008, International Standards Organisation, 2008.
- [10] Standards for processing aeronautical data. Standard RTCA/DO-200A, EUROCAE/ED-76, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, September 1998.
- [11] Commission Regulation (EU) No 73/2010 of 26 January 2010 laying down requirements on the quality of aeronautical data and aeronautical information for the single European sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:023:0006:0027:EN:PDF>, January 2010. Accessed: 27 January 2021.
- [12] Commission Implementing Regulation (EU) No 1207/2011 of 22 November 2011 laying down requirements for the performance and the interoperability of surveillance for the single European sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2011:305:0035:0052:EN:PDF>, November 2011. Accessed: 27 January 2021.
- [13] Information Technology — Security Techniques — Information Security Management Systems — Requirements. Standard BS ISO/IEC 27001:2013, International Standards Organisation, 2013.
- [14] Use of safety management systems by ATM service providers. Standard ESARR3, EUROCONTROL, 2011.

This page is intentionally blank

Index

- Accuracy Property, 17
- Artefact, Data, 6, 8, 13, 17, 18
- Assurance Level
 - Data, 7
- Completeness
 - Property, 17
- Consistency
 - Self, 16
 - With Inputs, 18
 - With Requirements, 16
- Cygnology
 - Black Swan Data, 10
 - Dragon King Data, 10
 - Perfect Storm Data, 10
 - Pudding Lane Data, 10
- Data
 - Entry, 17
 - DSAL, *see* Assurance Level, Data
 - Lifecycle
 - Data, 5
 - Engineering, 5
 - System, 5
 - Mitigation, 5, 6, 10
 - Property
 - Data, i, 5
 - Safety, 5
 - Safety Requirement
 - Data, 5, 7–9
 - Training
 - Personnel, 5

DATA IS HERE. DATA IS GROWING. DATA IS CAUSING HARM.

This book has been developed by the Safety-Critical Systems Club Data Safety Initiative Working Group (DSIWG) to provide guidance on how data, as distinct from hardware and software can be managed in a safety-related context.

"If you torture the data long enough, they confess – even to crimes that were never committed."

Nihat Bülent Gültekin

~

This is the seventh minor update since version 3.0. Paragraph numbering within the body of the document remains aligned with that major release. Thus users of any previous 3.x release of the guidance document will find migration to this edition takes little effort.

